



121/2

MATHEMATICS PAPER 2

MARKING SCHEME

1.	<table><tr><td>No</td><td>log</td></tr><tr><td>2.53^2</td><td>$2 \times 0.4031 = 0.8062$</td></tr><tr><td>83.45</td><td>1.9215</td></tr><tr><td>$\sqrt{0.4562}$</td><td>2.7277</td></tr><tr><td>790.9</td><td>$\frac{6592}{2}$ T.8296</td></tr><tr><td></td><td>$7.909 \times 10^2 = 2.8981$</td></tr></table>	No	log	2.53^2	$2 \times 0.4031 = 0.8062$	83.45	1.9215	$\sqrt{0.4562}$	2.7277	790.9	$\frac{6592}{2}$ T.8296		$7.909 \times 10^2 = 2.8981$		
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	$7.909 \times 10^2 = 2.8981$														
2.	$4x^2 - 3x - 6 = 0$ $x^2 - \frac{3}{4}x = \frac{6}{4}$ $\left(x - \frac{3}{8}\right)^2 = \frac{6}{4} + \frac{9}{64}$ $\left(x - \frac{3}{8}\right)^2 = \frac{105}{65}$ $x = \pm \sqrt{\frac{105}{64}} + \frac{3}{8}$ $x = 0.375 \pm 1.2889$ $x = 0.6639 \text{ or } -0.9139$														
3.	$T^2 = \frac{p^2 + h}{m^2} + R$ $T^3 - R = \frac{p^2 + n}{m^2}$ $p^2 = m^2(T^3 - R) - n$ $p = \pm \sqrt{m^2(T^3 - R) - n}$														
4. 1	$\frac{\sqrt{14\left(\sqrt{7+\sqrt{2}}\right)} - \sqrt{14\left(\sqrt{7-\sqrt{2}}\right)}}{7-2}$														



$$\frac{\sqrt{98 + \sqrt{28 - \sqrt{98 + \sqrt{28}}}}}{5}$$

$$2\sqrt{\frac{28}{5}}$$

$$\frac{2X2\sqrt{7}}{5}$$

$$\frac{4\sqrt{7}}{5}$$

$$a = \frac{4}{5} = b = 0$$

$$\begin{aligned} 5. \quad & 102^5(-x)^0 + 5.2^4(-x)^1 + 10.2^3(-x)^2 \\ & = 32 - 80x + 80x^2 \\ & (k=x)(32 - 80x + 80x^2) = -8x \\ & -80kx + 32x = -8x \\ & (-80k + 32)x = -8x \\ & -80k = -8 - 32 \\ & k = \frac{1}{2} \end{aligned}$$

$$6 \quad -\frac{2}{1}\begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix} + \frac{3}{1}\begin{pmatrix} x \\ y \\ 7 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 15 \end{pmatrix}$$

$$\begin{pmatrix} 3x \\ 3y \\ 37 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 15 \end{pmatrix} - \begin{pmatrix} -6 \\ -8 \\ 12 \end{pmatrix}$$

$$\begin{pmatrix} 3x \\ 3y \\ 37 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ 7 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

B is (2,3,1)

7.

2



$$5 - 4(1 - \sin^2 x) = 4 \sin x$$

$$5 - 4 + 4 \sin^2 x - 4 \sin x = 0$$

$$4 \sin^2 x - 4 \sin x + 1 = 0$$

$$4 \sin^2 x - 2 \sin x - 2 \sin x + 1 = 0$$

$$2 \sin x(2 \sin x - 1) - 1(2 \sin x - 1) = 0$$

$$(2 \sin x - 1)(2 \sin x - 1) = 0$$

$$\sin x = \frac{1}{2}$$

$$x = 30^\circ$$

8

$$\text{max mumArea} = 8.5 \times 5.5$$

$$= 46.75 \text{ cm}^2$$

$$\text{max imumarea} = 7.5 \times 4.5$$

$$= 33.75 \text{ cm}^2$$

$$\text{Rel.erro} = \frac{0.5}{8} + \frac{0.5}{5}$$

$$= 0.1625$$

$$A \text{ and } C \text{ in } 1 \text{ hr} = \frac{1}{45} + \frac{1}{30} = \frac{1}{8}$$

$$\text{in } 13 \text{ hrs} = \frac{13}{18}$$

$$\text{Remainder} = \frac{5}{18}$$

9

$$\text{if } B \text{ does } \frac{1}{40} \text{ in } 1 \text{ hr}$$

$$\frac{5}{18} = \frac{5}{18} \times \frac{40}{1}$$

$$= 11 \frac{2}{18}$$

$$= 11 \frac{1}{9} \text{ h}$$

10.	$x + 2y = -1$ $2x + 3y = 3$ $2x + 4y = -2$ $2x + 3y = 3$ $y = -5$ $x = -9$ $\text{centre}(-9, -5)$ $(x + 9)^2 + (y + 5)^2 = 5^2$ $x^2 + 18x + 81 + y^2 + 10y + 25 = 25$ $x^2 + y^2 + 18x + 10y + 81 = 0$		
11.	$\sqrt{18^2 - 5^2}$ $= 17.292\text{cm}$ $\text{NC} = \frac{\sin \alpha}{17.292} = \frac{5}{17.292}$ $\alpha = 16.81^\circ$ $< \text{MUN} = 2 \times 16.81$ $= 33.62^\circ$		
12.	$\frac{dy}{dx} = 3x^2 - 4x + 3$ $3(2)^2 - 4(2) + 3 = 7$ $y = mx + c$ $M_2 = -\frac{1}{7}, y = -\frac{1}{7}x + C, C = 37$ $y = -\frac{1}{7x} + 37(\text{normal})$		



13. $\sqrt{8^2} - 3^2$
 $7.4cm$

$$p = KQ^3 + \frac{m}{Q^2}$$

$$8k + \frac{m}{4} = 108$$

$$32K + M = 432$$

$$27k + \frac{m}{9} = 259$$

14. $243K + m = 2331$

$$211K = 1899$$

$$K = 9, m = 144$$

$$P = 9x6^3 + \frac{144}{6^2}$$

$$= 1948$$

$${}_4y + \frac{1}{\log_4} y = 2$$

$$\text{let } x = \log_4 y$$

$$x + \frac{1}{x} = 2$$

$$x^2 + 1 = 2x$$

$$2^2 - 2x + x + 1 = 0$$

15. $\log(x-1)(x-1) = 0$

$$x = 1 \text{ or } x = 1$$

$$x = \log_4 y$$

$$x = 1$$

$$Y = 4$$



$$16. \quad \text{M.A.D.} = \sum \left(\frac{X - \bar{X}}{N} \right)$$

$$\bar{X} = \frac{\sum X}{N} = \frac{44 + 32 + 71 + 52 + 28 + 39 + 46 + 64}{8}$$

$$= \frac{376}{8}$$

$$= 47$$

$$M.A.D = \frac{92}{8}$$

$$= 11.5$$

17. (a). Taxable income $30,000 + 10480 = 40,480$

In \$=2024

$$435 \times 2 = 870$$

$$435 \times 3 = 1305$$

$$435 \times 4 = 1740$$

$$435 \times 5 = 2175$$

$$284 \times 6 = 1704$$

$$\text{Gross tax} = 7794$$

(b). Net tax = $7794 - 800$

$$= 6994$$

(c). $870 + 1305 + 1740 + 2175 + (1296 \times 6)$



$$\text{New net tax} = 13866 - 800$$

$$= 13066$$

$$\text{Increase in tax} = 13066 - 6994$$

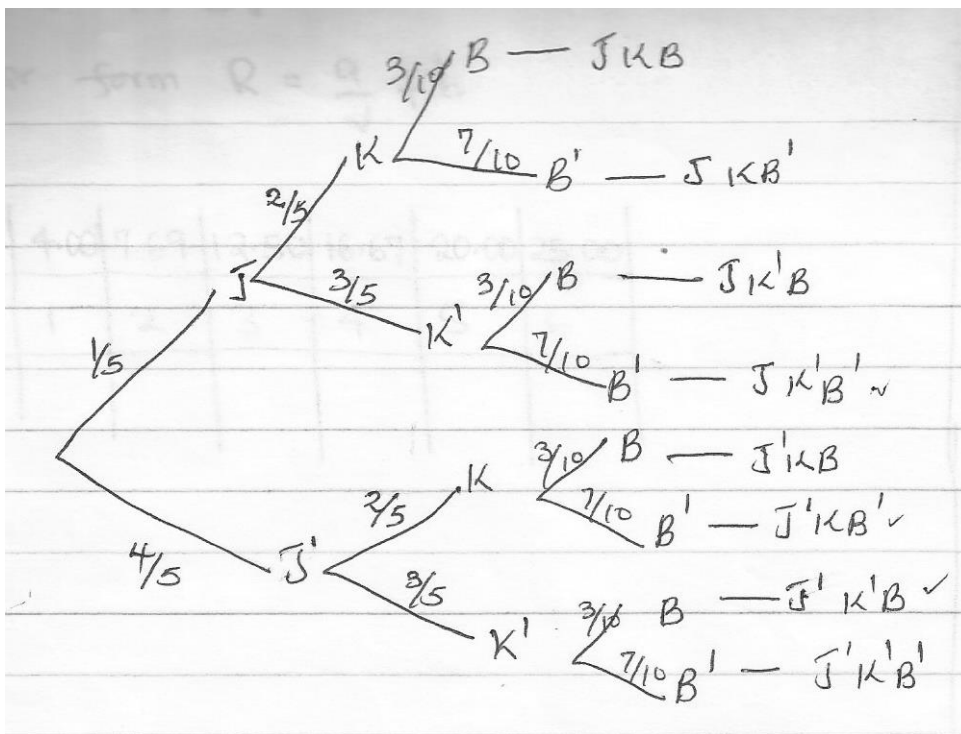
$$= 6072$$

$$\begin{aligned} \text{\%age increase} &= \frac{6072}{6994} \times 100 \\ &= 86.82\% \end{aligned}$$

18.

No.	Working
	(a). Calculate (i) Value of x Distance between A to B = $600 \times 3.5 = 2100\text{nm}$ $60\theta = 2100$ $\theta = \frac{2100}{60} = 35^\circ$ $x = 35^\circ - 14^\circ = 21^\circ$ (ii). Distance between Band C $\frac{90}{360} \times 3.142 \times 2 \times 6370$ $BC = 100,0072.7\text{km}$ (b). D is an airport situated at $(5^\circ \text{N}, 120^\circ \text{W})$ (i). shortest Distance between C and D $CD = 164 \times 60 = 9840\text{nm}$ Time = $\frac{9,840}{60}$ Time = 16 hrs 24min (ii). Local time at D Change in Longitude between A and D = $120^\circ - 30^\circ = 90^\circ$ $90^\circ \Rightarrow 4 \text{ min}$ $190^\circ \Rightarrow 90^\circ \times 4 = \frac{360}{60} 6\text{hrs}$ Time at D; $12.20 \text{ p.m} - 6\text{hrs} = 6.20\text{am}$

19. (a).



(JorB)

 $P(JK^1B^1) \text{ or } P(J^1K^1B)$

$$(b).(i). \quad P\left(\frac{1}{5} \times \frac{3}{5} \times \frac{7}{10}\right) + \left(\frac{4}{5} \times \frac{3}{5} \times \frac{3}{10}\right)$$

$$\frac{21}{250} + \frac{36}{250} = \frac{57}{250}$$

$$(ii). \quad P(\text{All fair}) = \frac{4}{5} \times \frac{3}{5} \times \frac{7}{10} = \frac{42}{125}$$

(only two fail)

$$(iii). \quad P\left(\frac{1}{5} \times \frac{3}{5} \times \frac{7}{10}\right) + \left(\frac{4}{5} \times \frac{2}{5} \times \frac{7}{10}\right) + \left(\frac{4}{5} \times \frac{3}{5} \times \frac{3}{10}\right)$$

$$\frac{21}{250} + \frac{56}{250} + \frac{36}{250} = \frac{113}{250}$$

20. (a). R.V = a + bv



Linear form $R = \frac{a}{v} + b$

(b).

$$(10.5, 2.5)(22.5, 5.4)$$

$$(c). (i). a = \frac{5.4 - 2.5}{22.5 - 10.5}$$

$$= 0.2417$$

$$(ii). b = 0$$

$$\frac{a + 6d}{3 + 3d} = \frac{a + 14d}{a + 6d}$$

$$a^2 + 12ad + 36d^2 = a^2 + 15ad + ad + 45d^2$$

$$12ad + 36d^2 = 18ad + 45d^2$$

21.

$$(a). -6a = ad \dots\dots\dots(i)$$

$$sn6 = 3(2a + 5d) = 12$$

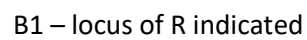
$$2a + 5d = 4 \dots\dots\dots(ii)$$

$$a = -3, d = 2$$

$$(b) \frac{a + 6d}{a + 3d} = \frac{a}{3}$$

$$= 3$$

22.



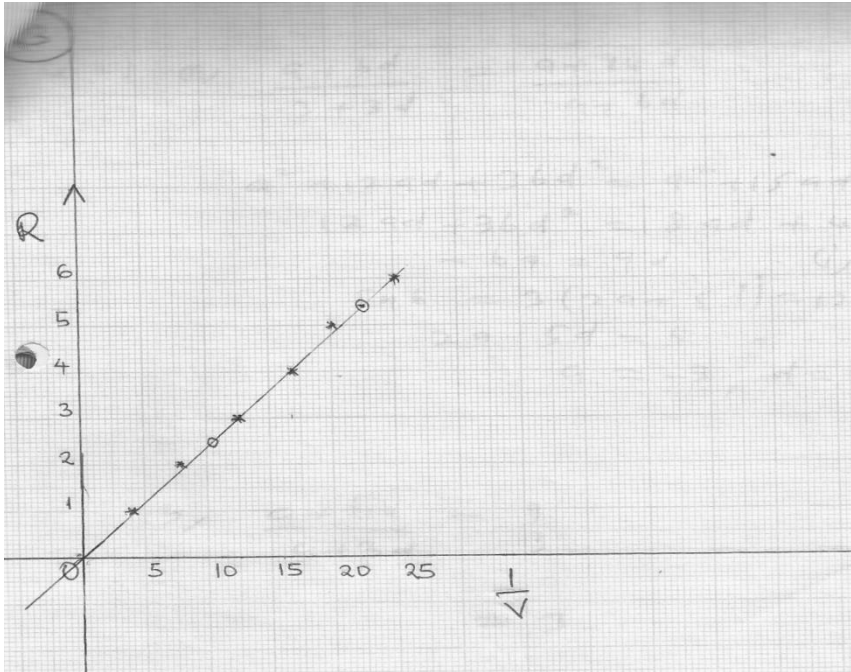
10

$$80x + 50y \leq 400$$

$$x > 0$$

$$y > 0$$

(b).



(c). 2 cakes of type A

3 cakes of type B



$$\frac{ds}{dt} = 30 - 10t = 0 \dots\dots\dots$$

$$t = 3 \text{ sec} \dots\dots\dots$$

$$\begin{aligned} 24. (a). \text{ maximum height} &= 30(3) - 5(3)^2 \dots\dots\dots \\ &= 90 - 45 \\ &= 45 \dots\dots\dots \end{aligned}$$

Time reached is 3sec

(b). At original level $S=0$

$$0 = 30t - 5t^2 \dots\dots\dots$$

$$t(30 - 5t) = 0$$

$$t = 0 \text{ or } t = 6 \dots\dots\dots$$

(c).

$$\begin{aligned} (c). \quad v &= \frac{ds}{dt} = 30 - 10t \dots\dots\dots \\ \text{at } t = 4, 30 - 40 &= -10 \text{ m/s} \dots\dots\dots \end{aligned}$$

(d). $a = \frac{dv}{dt} = -10 \dots\dots\dots$

Hen acceleration = $-10 \text{ m/s}^2 \dots\dots\dots$

